CS445 / SE463 / CS645
Software Requirements Specification and Analysis
Spring 2026

Deliverable 2: Domain Model & Stakeholders

Group Name: The Reservists
System Name: Carrel

Kevin Wang
Chris Wu
Rui Lin

June 5, 2026

1 Project Identification

Item	Description
Group Name	The Reservists
System Name	Carrel
Project Name	Carrel: A Campus Study Space Booking System
Primary Product Perspective	A new campus-wide study space booking system, not a successor to a specific existing system.

Carrel is a campus study space booking system designed to help students find, reserve, check in to, and cancel study space bookings. The system also supports administrators who manage room inventory, booking rules, maintenance closures, no-show policies, disputes, and administrative overrides. The main goal is to make study space allocation easier, fairer, and less wasteful by reducing ghost reservations and centralizing booking information.

2 Stakeholders

A stakeholder is any person, group, organization, or external system that is involved in the project, affected by the system's process or outcome, or able to influence the system's success. Table 1 lists the stakeholders for Carrel. Similar end users are grouped together as one stakeholder type.

Table 1: Stakeholders for Carrel

Category	Stakeholder	Relationship to Carrel	Benefit Obtained
Primary Users			
Primary Users	Students	Students are the primary end users. They search for available study spaces, filter spaces by building, time, capacity, and amenities, reserve slots, check in on arrival, and cancel future bookings.	They benefit from a centralized and fairer way to find study space. They also benefit from fewer ghost reservations because unused rooms can return to the available pool after the grace window.
Primary Users	Student Government and Student Representatives	Student representatives advocate for student interests and may provide feedback on quota rules, no-show penalties, dispute processes, accessibility concerns, and usability problems.	They benefit from a more transparent system for study-space allocation and from usage evidence that can support requests for better student services or more study space.
Administrative Stakeholders			

Category	Stakeholder	Relationship to Carrel	Benefit Obtained
Administrative Stakeholders	Facilities Administrators	Facilities administrators manage study space inventory and booking rules. They add spaces, retire rooms, edit amenities, schedule maintenance closures, configure policies, review disputes, and apply overrides.	They benefit from better control over campus study-space usage, fewer manual booking conflicts, and a clear audit trail for resolving disputes.
Administrative Stakeholders	Library Administration Staff	Library staff are involved because many study rooms, individual carrels, and quiet study spaces are located in libraries. They may help define local room rules and monitor library study-space usage.	They benefit from replacing paper sign-up sheets and informal room claims with a centralized booking record, reducing room conflicts and making library space usage easier to manage.
Administrative Stakeholders	Academic Departments and Room Owners	Some study rooms or project spaces may belong to academic departments, faculties, or other room-owning units. These stakeholders care about how their rooms are made bookable and when they are blocked for departmental use.	They benefit from a controlled mechanism for sharing spaces with students while preserving administrative control over departmental rooms and special-use spaces.
Campus Operations			
Campus Operations	Campus IT / Identity and Access Management Team	The campus IT team supports authentication and integration with university identity services. Carrel depends on the identity service to verify that a user is a valid student or administrator.	IT benefits because Carrel can rely on centralized identity management instead of creating separate accounts, passwords, and user-management processes.
Campus Operations	Accessibility Services	Accessibility services are affected because students may need study spaces with specific accessibility features, such as wheelchair access, quiet rooms, accessible furniture, or proximity to elevators.	They benefit if Carrel records accessibility-related amenities clearly and makes accessible spaces easier for students to find and reserve.
Campus Operations	Campus Security and Building Access Staff	Campus security or building access staff may be involved when a booking dispute, after-hours access issue, unauthorized occupancy issue, or room misuse incident occurs.	They benefit from an audit trail showing who booked a room, when the booking was scheduled, whether the student checked in, and whether an administrator applied an override.

Category	Stakeholder	Relationship to Carrel	Benefit Obtained
Campus Operations	Cleaning and Maintenance Staff	Cleaning and maintenance staff are affected by room availability, maintenance closures, repair windows, and high-demand periods. Carrel can mark spaces as unavailable during cleaning, maintenance, or repair.	They benefit from clearer scheduling and fewer conflicts between room usage and operational maintenance work.
Institutional Stakeholders			
Institutional Stakeholders	Registrar / Exam Scheduling Office	The registrar or exam scheduling office may need to block study spaces during exam-period administration, special events, or other campus scheduling needs.	They benefit from blackout-period rules that prevent normal student bookings from conflicting with institutional scheduling priorities.
Institutional Stakeholders	University Leadership and Space Planning Staff	University leadership and space planning staff are interested in long-term usage data, peak-demand periods, under-utilized spaces, and mismatches between study-space supply and student demand.	They benefit from Carrel's usage records because those records can support future space planning, facility investment, and resource allocation decisions.
External Interface Stakeholders			
External Interfaces	External Calendar Service Provider	An external calendar service receives booking event information from Carrel when a booking is confirmed.	The calendar service benefits from structured event information, while students benefit by seeing the booking time, building, room number, and cancellation link in their own calendar.
External Interfaces	University Identity Service	The university identity service is an external interface used by Carrel to authenticate students and administrators through their institutional accounts.	It benefits by remaining the authoritative source for user identity, while Carrel benefits by avoiding duplicate identity records and reducing incorrect user access.

3 Domain Model

Figure 1 shows the domain model for Carrel. The model focuses on the problem-world concepts relevant to study space booking rather than implementation classes. It includes active agents such as students and administrators, physical and conceptual entities such as buildings, study spaces,

bookings, policies, and disputes, and interface/shared-phenomenon concepts such as check-in codes, QR codes, identity service authentication, and calendar invites.

The diagram uses the following conventions:

- `<<actor>>` marks active external agents that interact with or influence the system.
- `<<interface>>` marks external services or systems that Carrel interacts with.
- `<<shared phen.>>` marks phenomena visible to both the environment and Carrel.
- Filled diamonds represent composition, where the contained concept is strongly owned by the whole.
- Hollow diamonds represent aggregation, where a concept is associated with a whole but may remain conceptually separate.
- Multiplicities appear on associations to show how many instances may participate in each relationship.

Figure 1. Carrel Domain Model

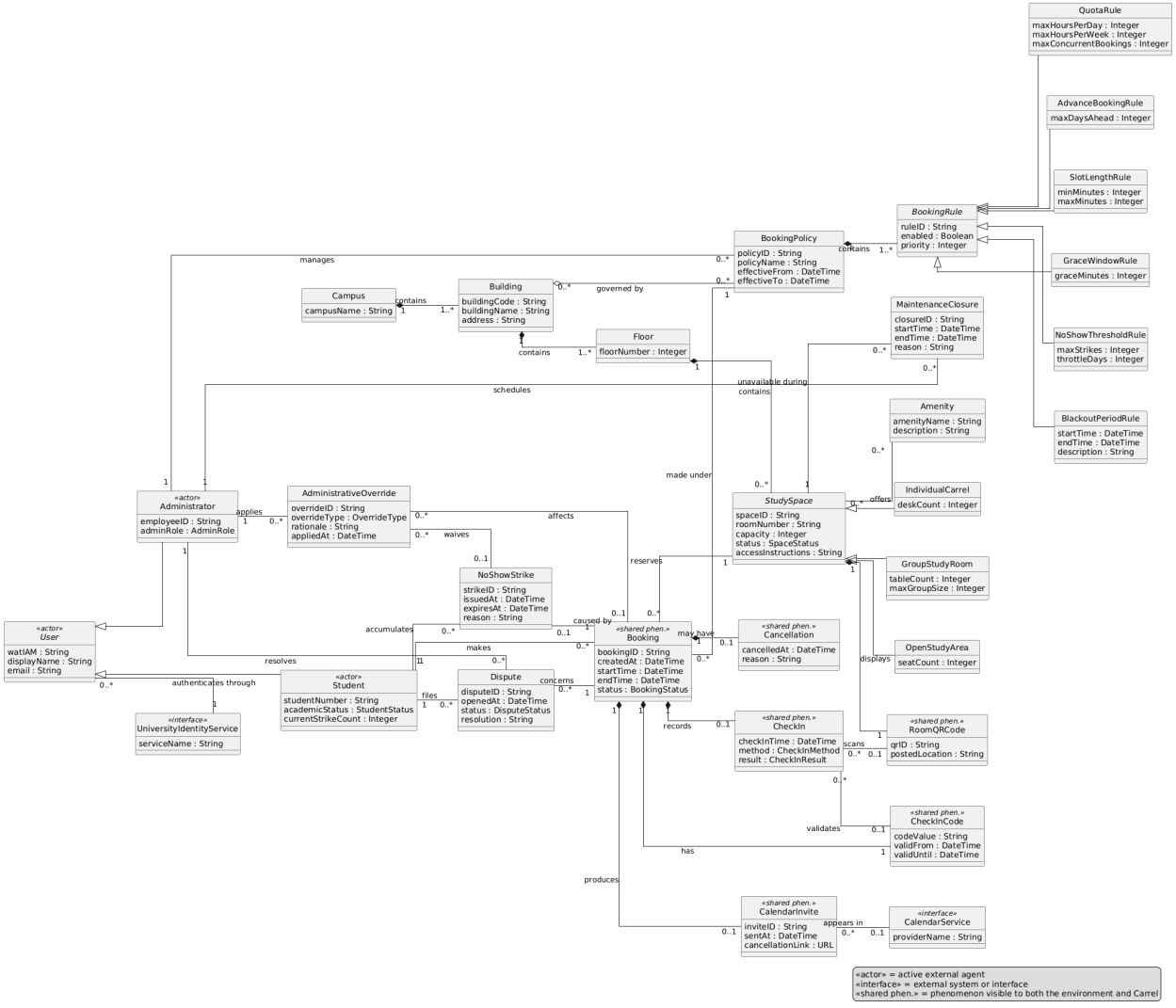


Figure 1: Carrel Domain Model

4 Domain Model Coverage Notes

The domain model is designed to cover the major concepts and relationships needed for Carrel's booking, check-in, cancellation, rule-management, and dispute-resolution responsibilities.

4.1 Conceptual Objects

The main conceptual objects include:

- **User, Student, and Administrator**, representing the main human agents.
- **Campus, Building, Floor, and StudySpace**, representing the physical study-space inventory.
- **IndividualCarrel, GroupStudyRoom, and OpenStudyArea**, representing specialized kinds of study spaces.
- **Booking, CheckInCode, CheckIn, Cancellation, CalendarInvite, NoShowStrike, and Dispute**, representing the booking lifecycle and related records.
- **BookingPolicy and BookingRule**, representing configurable booking governance.
- **QuotaRule, AdvanceBookingRule, SlotLengthRule, GraceWindowRule, NoShowThresholdRule, and BlackoutPeriodRule**, representing specialized booking rules.
- **MaintenanceClosure and AdministrativeOverride**, representing administrative control over room availability and enforcement outcomes.
- **UniversityIdentityService and CalendarService**, representing external interfaces.

4.2 Important Associations and Multiplicities

The diagram includes multiplicities on the important associations. For example, one student may make zero or more bookings, while each booking is made by one student. A booking reserves exactly one study space, while a study space may be associated with zero or more bookings over time. A booking may have a check-in, cancellation, calendar invite, no-show strike, or dispute depending on the lifecycle outcome.

4.3 Composition, Aggregation, and Specialization

The model uses composition for strong containment relationships such as a campus containing buildings, a building containing floors, and a floor containing study spaces. It uses aggregation for relationships such as buildings being governed by booking policies, because a policy can be conceptually associated with multiple building contexts. The model also uses specialization and generalization: *Student* and *Administrator* specialize *User*; *IndividualCarrel*, *GroupStudyRoom*, and *OpenStudyArea* specialize *StudySpace*; and the different rule classes specialize *BookingRule*.

5 Conclusion

This D2 report identifies the group and system name, lists more than ten stakeholders with their relationship to Carrel, and provides a UML domain model for the study space booking domain. The model captures the major real-world concepts involved in reserving spaces, enforcing check-in and cancellation rules, managing administrative policies, handling maintenance closures, and resolving disputes.